



OPEN Validation of histopathology foundation models through whole slide image retrieval

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We evaluated several foundation models in histopathology for image retrieval using a zero-shot approach. These models generated embeddings that were directly employed for retrieval without additional fine-tuning. Our experiments were conducted on diagnostic slides from The Cancer Genome Atlas (TCGA), which covers 23 organs and 117 cancer subtypes. We used Yottixel as the framework for whole-slide image (WSI) retrieval via patch-based embeddings. Retrieval performance was evaluated using macro-averaged F1 scores for top-1, top-3, and top-5 retrievals. The top-5 retrieval F1 scores indicated varying levels of performance: Yottixel-DenseNet (27% $\pm$ 13%), Yottixel-UNI (42% $\pm$ 14%), Yottixel-Virchow (40% $\pm$ 13%), Yottixel-GigaPath (41% $\pm$ 13%), and GigaPath WSI (40% $\pm$ 14%). These results demonstrate the potential and limitations of foundation models for histopathology image retrieval, underscoring the need for further advancements in embedding and retrieval techniques.

The emergence of digital pathology and artificial intelligence (AI) seems to bring about an eminent revolution in the field of histopathology, with anticipated efficiency and accuracy at unprecedented levels for diagnostic and treatment planning of many diseases. Digital pathology offers high-resolution tissue images that can be processed and analyzed by advanced computer algorithms. This shift could enable pathologists to leverage the power of deep learning models to identify tissue patterns, classify diseases, and predict patient outcomes with greater precision. In this context, image retrieval may also play a critical role in enhancing diagnostic workflows, provided we can finally close the semantic gap after three decades of research. Image search and retrieval in histopathology is particularly significant as it allows pathologists to search databases - small or large - of digital slides to find visually, i.e., anatomically, similar cases¹. This capability is invaluable for comparative diagnosis, where finding similar cases can aid in confirming or refining a diagnosis. However, the effectiveness of image retrieval depends, among others, on the quality of embeddings produced by deep neural networks². High-quality embeddings ensure that similar images are clustered closely together in the feature space, leading to more accurate retrieval results. Conversely, poor-quality embeddings can result in misclassification and retrieval of irrelevant cases, potentially compromising diagnostic accuracy. The advent of foundation models, large-scale AI models pre-trained on vast amounts of data, holds great promise in addressing the long-standing semantic gap problem in image search and retrieval for histopathology³. The *semantic gap* refers to the disparity between the high-level concepts understood by human experts and the low-level features captured by machines. Foundation models, with their expected ability to generalize across diverse tasks and domains, could bridge this gap by learning more robust and semantically meaningful representations of medical images. This development offers hope that AI-driven image retrieval will become more accurate and reliable, ultimately leading to better patient outcomes and more efficient healthcare systems. However, despite the potential of foundation models, their application in zero-shot WSI retrieval within histopathology remains underexplored. This paper seeks to fill this gap by evaluating several foundation models for zero-shot WSI retrieval, using a diverse dataset from The Cancer Genome Atlas (TCGA). The findings aim to provide insights into the effectiveness of these models in clinical-grade image retrieval, identifying key areas for improvement in embedding techniques and retrieval methodologies.

Results

The zero-shot retrieval performance of several foundation models was evaluated on whole-slide images (WSIs) using the Yottixel platform. The models include Yottixel-DenseNet (default), Yottixel-UNI, Yottixel-Virchow,

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	Top-1	Majority @ Top-3	Majority @ Top-5
Yottixel	28% ± 13% [23% 33%]	28% ± 13% [17% 32%]	27% ± 13% [22% 32%]
Yottixel-UNI	44% ± 15% [38% 50%]	44% ± 14% [38% 50%]	42% ± 14% [36% 48%]
Yottixel-Virchow	41% ± 15% [35% 47%]	41% ± 14% [35% 47%]	40% ± 13% [36% 45%]
Yottixel-GigaPath	43% ± 15% [37% 49%]	43% ± 15% [37% 49%]	41% ± 13% [35% 47%]
GigaPath WSI	42% ± 15% [36% 48%]	41% ± 15% [35% 47%]	40% ± 14% [35% 45%]

Table 1. The macro average and 95% confidence interval of F1 scores for tested WSI retrievals. Yottixel, in its default configuration using pre-trained DenseNet, serves as the testbed and benchmark. DenseNet was replaced by UNI, Virchow and GigaPath to provide Yottixel with patch embeddings [see Fig. 1]. WSI retrievals were performed using Yottixel’s median-of-minimums scheme. GigaPath WSI (last row) is the only results independent from Yottixel as GigaPath can provide one embedding for WSI through aggregation of all patch embeddings.

Organ	Yottixel	Yottixel+UNI	Yottixel+Virchow	Yottixel+GigaPath	GigaPath-WS
Adrenal glands	0.24	0.42	0.37	0.40	0.38
Bladder	0.53	0.66	0.65	0.63	0.67
Brain	0.26	0.39	0.44	0.42	0.41
Breast	0.17	0.27	0.25	0.26	0.25
Cervix	0.15	0.27	0.27	0.31	0.31
Colon	0.54	0.61	0.58	0.60	0.54
Esophagus	0.41	0.63	0.61	0.78	0.71
Eye	0.20	0.41	0.37	0.42	0.34
Head and neck	0.23	0.44	0.37	0.46	0.35
Kidneys	0.50	0.82	0.78	0.71	0.73
Lining	0.37	0.35	0.32	0.31	0.30
Liver	0.33	0.33	0.33	0.33	0.33
Lungs	0.12	0.21	0.21	0.21	0.21
Pancreas	0.26	0.50	0.47	0.50	0.5
Prostate	0.50	0.50	0.50	0.50	0.5
Rectum	0.26	0.66	0.55	0.63	0.57
Skin	0.20	0.30	0.22	0.26	0.24
Soft tissues	0.18	0.43	0.37	0.42	0.39
Stomach	0.14	0.30	0.29	0.30	0.34
Testicles	0.23	0.35	0.36	0.35	0.34
Thymus	0.11	0.29	0.25	0.26	0.27
Thyroid gland	0.32	0.47	0.41	0.42	0.41
Uterus	0.26	0.52	0.45	0.49	0.52

Table 2. Macro average of F1 score for top-1 WSI retrievals. Best results highlighted in GREEN for each organ. PINK highlighted results show GigaPath results where aggression has worsened the patch-level results.

Yottixel-GigaPath, and GigaPath WSI, with retrieval accuracy measured using macro-averaged F1 scores for top-1, top-3, and top-5 retrievals across 23 organs and 117 cancer subtypes. Table 1 shows the overall results for all WSI retrievals, whereas Tables 2 and 3, and 4 show the detailed results for top-1, majority of top-3 and majority of top-5 macro average F1 scores, respectively.

Organ	Yottixel	Yottixel+UNI	Yottixel+Virchow	Yottixel+GigaPath	GigaPath-WS
Adrenal glands	0.25	0.43	0.38	0.41	0.31
Bladder	0.50	0.68	0.66	0.66	0.66
Brain	0.28	0.43	0.44	0.42	0.43
Breast	0.16	0.30	0.23	0.25	0.25
Cervix	0.12	0.30	0.30	0.35	0.33
Colon	0.50	0.58	0.57	0.58	0.51
Esophagus	0.37	0.61	0.60	0.76	0.69
Eye	0.20	0.37	0.39	0.43	0.42
Head and neck	0.23	0.43	0.30	0.43	0.38
Kidneys	0.50	0.78	0.77	0.70	0.73
Lining	0.38	0.35	0.32	0.37	0.26
Liver	0.33	0.33	0.33	0.33	0.33
Lungs	0.11	0.22	0.23	0.22	0.20
Pancreas	0.22	0.47	0.50	0.50	0.48
Prostate	0.50	0.5	0.50	0.50	0.50
Rectum	0.28	0.57	0.52	0.53	0.50
Skin	0.21	0.30	0.23	0.24	0.21
Soft tissues	0.18	0.45	0.38	0.42	0.36
Stomach	0.15	0.35	0.32	0.30	0.32
Testicles	0.26	0.38	0.37	0.34	0.37
Thymus	0.11	0.32	0.29	0.28	0.31
Thyroid gland	0.32	0.47	0.42	0.43	0.43
Uterus	0.26	0.52	0.44	0.52	0.52

Table 3. Macro average of F1 score for majority of top-3 WSI retrievals. Best results highlighted in GREEN for each organ. PINK highlighted results show GigaPath results where aggression has worsened the patch-level results.

Overall retrieval performance

Yottixel-DenseNet, the baseline model, achieved the lowest performance, with an F1 score of $27\% \pm 13\%$ for top-5 retrievals. The more advanced foundation models, such as Yottixel-UNI and Yottixel-GigaPath, showed improved retrieval accuracy. For top-5 retrievals, Yottixel-UNI achieved a $42\% \pm 14\%$ F1 score, while Yottixel-GigaPath achieved $41\% \pm 13\%$. Despite these improvements, the overall retrieval accuracy across models remains below 50%, reflecting the difficulty of zero-shot retrieval in complex histopathological datasets.

Organ-specific performance

The retrieval performance varied significantly across different organs. Organs such as the bladder, kidneys, and esophagus showed notably higher retrieval accuracy. For instance, Yottixel-UNI achieved an F1 score of 82% for kidney tissues, and Yottixel-GigaPath achieved 78%. These higher scores reflect the relative ease of capturing homogeneous and well-structured tissue patterns in these organs.

In contrast, the retrieval performance for organs such as the lungs and cervix was much lower. For lung tissues, all models struggled, with Yottixel-UNI achieving a top-1 F1 score of just 21%. This likely reflects the intrinsic heterogeneity of lung tissue, where patch-based embeddings may fail to capture the complexity of the tissue structure.

GigaPath aggregation

A notable aspect of the study was the comparison between patch-based embeddings and the aggregation method used by GigaPath WSI. While aggregation theoretically reduces computational complexity, GigaPath WSI did not lead to substantial performance improvements and, in some cases, performed worse than patch-based models. For example, for adrenal glands, GigaPath WSI achieved 38%, whereas Yottixel-GigaPath achieved 40%, suggesting that the aggregation process might lose important spatial information.

Top-3 and Top-5 retrievals

Across most models, the retrieval accuracy improved slightly when expanding from top-1 to top-3 and top-5 retrievals. For example, Yottixel-UNI's F1 score for kidney tissues increased from 68% for top-1 retrievals to 76% for top-5 retrievals, indicating that the model is capable of retrieving multiple relevant cases when allowed

Organ	Yottixel	Yottixel+UNI	Yottixel+Virchow	Yottixel+GigaPath	GigaPath-WSI
Adrenal glands	0.25	0.36	0.35	0.34	0.29
Bladder	0.48	0.70	0.63	0.69	0.65
Brain	0.29	0.44	0.43	0.45	0.43
Breast	0.15	0.31	0.24	0.26	0.27
Cervix	0.11	0.23	0.29	0.28	0.30
Colon	0.46	0.56	0.54	0.55	0.56
Esophagus	0.42	0.61	0.59	0.61	0.58
Eye	0.27	0.36	0.36	0.35	0.36
Head and neck	0.23	0.28	0.29	0.39	0.3t1
Kidneys	0.52	0.76	0.68	0.67	0.71
Lining	0.27	0.38	0.37	0.37	0.30
Liver	0.33	0.33	0.33	0.33	0.33
Lungs	0.11	0.20	0.21	0.18	0.18
Pancreas	0.23	0.47	0.45	0.45	0.40
Prostate	0.50	0.50	0.50	0.50	0.50
Rectum	0.26	0.56	0.37	0.53	0.46
Skin	0.19	0.23	0.23	0.24	0.19
Soft tissues	0.19	0.46	0.37	0.40	0.37
Stomach	0.14	0.35	0.29	0.30	0.33
Testicles	0.25	0.41	0.39	0.36	0.37
Thymus	0.10	0.36	0.32	0.29	0.37
Thyroid gland	0.33	0.44	0.44	0.43	0.45
Uterus	0.21	0.46	0.43	0.45	0.48

Table 4. Macro average of F1 score for majority of top-5 WSI retrievals. Best results highlighted in GREEN for each organ. PINK highlighted results show GigaPath results where aggression has worsened the patch-level results.

a broader selection window. However, for more challenging organs, such as the lungs, the improvements were marginal, with little increase in accuracy even when considering more top-ranked results.

Methods

Data

We used diagnostic slides from The Cancer Genome Atlas (TCGA), consisting of 11,444 WSIs from 9,339 patients, spanning 23 organs and 117 cancer subtypes. Detailed information on all subtypes for each organ, including the number of patients and WSIs, can be found in Tables TA1 to TA23 in the Appendix. TCGA projects, such as lung adenocarcinoma (LUAD) and lung squamous cell carcinoma (LUSC), are broad classifications covering approximately 98% of cases. However, identifying specific subtypes within these broader categories can provide more detailed insights into tumor biology, prognosis, and treatment. For example, lepidic adenocarcinoma (formerly bronchioalveolar) is often treated successfully with surgery alone, whereas adenocarcinoma with mixed subtypes is associated with a higher likelihood of metastasis and worse clinical outcomes. While most studies focus on these generic tumor types, detailed subtyping, as we have included in our study, is crucial for more accurate retrieval and better clinical decision-making⁴.

Search engine

To evaluate the embeddings of the foundation models in a zero-shot manner, we selected a dependable search framework that met several essential criteria. First, the framework needed to allow easy integration of new models without disrupting its overall design, requiring minimal additional ablation studies. Second, it was necessary for the framework to have an unsupervised patching algorithm that could framework patch-based searches. Third, the framework should support whole-slide image (WSI) retrievals by utilizing all selected patches. Lastly, the framework had to have a proven record of storage efficiency, search speed, and high accuracy compared to other systems. After considering these factors, we chose Yottixel as our testing platform^{1,2,5}. Yottixel's flexible topology, as shown in Fig. 1, allows for seamless integration of any deep learning model. This made it an ideal choice for our zero-shot retrieval tasks, as it maintained its structural integrity while accommodating the new models.

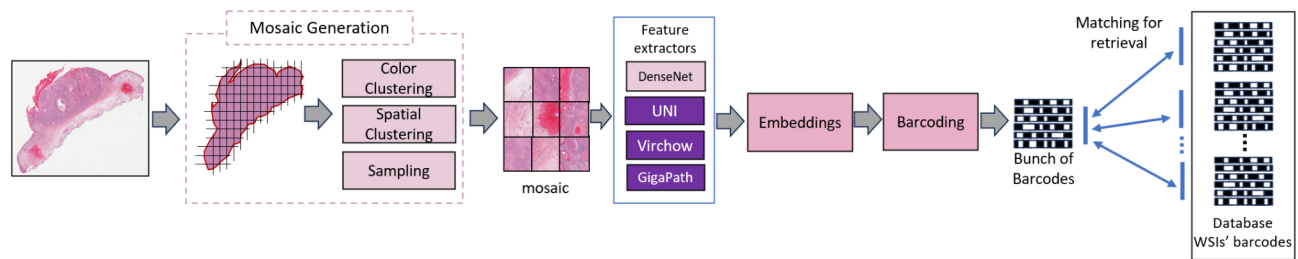


Fig. 1. Framework Overview: A comprehensive pipeline for histopathological image processing and retrieval. The process begins with an input histopathology slide image (leftmost) that undergoes Mosaic Generation through three parallel processes: Color Clustering, Spatial Clustering, and Sampling, resulting in a mosaic representation of the tissue. The mosaic patches are then processed through multiple feature extractors (DenseNet, UNI, Virchow, and GigaPath) to generate rich feature representations. These features are transformed into embeddings, which are then converted into compact barcodes. These barcodes form a bunch of barcode representations for the input image, which are then matched against a database of WSI (Whole Slide Image) barcodes using medium-of-minimum Hamming distance calculations for efficient retrieval and comparison⁵. This matching strategy helps identify the most similar cases in the database by computing distances between the query barcodes and the indexed WSI barcodes. This architecture enables effective content-based retrieval of histopathological images by transforming complex visual data into compact, comparable representations while maintaining retrieval accuracy through sophisticated distance metrics.

Patching WSIs

For patching WSIs, we employed Yottixel’s “mosaic” patching method, which operates in two unsupervised stages. In the first stage, we segmented the WSI into distinct regions based on color composition using k-means clustering. This segmentation allowed us to capture the pattern variability across different regions within the slide, including structures such as blood, muscle tissue, and fat. Based on visual inspection across various WSIs, we typically set the number of color clusters to nine⁵. In the second stage, we selected a small percentage of representative patches from each color-segmented region to preserve spatial diversity. This selection process used k-means clustering once again, grouping patches based on their location within each color region.

Unlike the default Yottixel settings, which use 1024×1024 patches, we extracted the patches in 224×224 pixels (20x magnification with 0.5 microns per pixel (mpp)) to accommodate the input requirements of the foundation models. Additionally, we reduced the number of representative patches to 2% (instead of the default 5%) to keep the patch count manageable without sacrificing key information. This “mosaic” approach significantly reduces computational complexity while retaining the most critical visual patterns and structures in the WSI. By using this compact mosaic of patches, we minimized the processing burden of the gigapixel WSI, ensuring efficient yet comprehensive representation of the slide.

Evaluation metric

We evaluated the models using the macro-average F1-score for top-1, majority top-3, and majority top-5 WSI retrievals. Since the dataset is imbalanced, using the macro-average provides a more rigorous and reliable evaluation. By weighting each class equally, regardless of its prevalence, this metric ensures that the evaluation reflects performance across all classes, not just those with the most samples.

Models for search engine

We evaluated three foundation models (i.e., UNI, Virchow, and GigaPath) each pretrained on large-scale pathology datasets using self-supervised learning techniques. These models leverage vision transformer (ViT) architectures and are designed to generate embeddings suitable for histopathology image retrieval.

UNI is a self-supervised vision encoder based on a Vision Transformer Large (ViT-L) architecture⁶. It was pretrained on the Mass-100 K dataset, comprising over 100 million patches from 100,426 Hematoxylin and Eosin (H&E) stained slides across 20 major tissue types. The pretraining process utilized DINOv2, a self-supervised learning framework that combines student-teacher knowledge distillation with masked image modeling. UNI was optimized using two main loss functions: self-distillation loss and masked image modeling loss. The dataset used for training included 100,130,900 image patches at varying resolutions (i.e., 75,832,905 patches at 256×256 pixels and 24,297,995 patches at 512×512 pixels) all at 20x magnification (0.5 mpp). This extensive training allows UNI to generate robust embeddings for tissue patches, making it well-suited for patch-based image retrieval tasks.

Virchow is a self-supervised vision transformer model specifically developed for computational pathology⁷. The model uses a Vision Transformer Huge (ViT-H/14) architecture with 632 million parameters. Pretraining was conducted on a large dataset of approximately 1.5 million H&E-stained whole-slide images (WSIs) from 100,000 patients, covering 17 distinct tissue types, including both cancerous and benign specimens. Tissue samples were collected from biopsies (63%) and resections (37%). Similar to UNI, Virchow employs the DINOv2 framework for self-supervised learning, which enables the model to learn meaningful tissue representations without labeled data. Virchow processes 224×224 pixel tiles extracted from WSIs at 20x magnification (0.5 mpp), and its training involved around 2 billion image tiles drawn from a total pool of 13 billion available tiles.

	Size	FLOPS	Embed.Size	Storage (BoB))
DensNet	6,953,856	2.83 G	270×1024	1500 MB
UNI	303,350,784	59.38 G	270×1024	1500 MB
Virchow	631,229,184	162.05 G	270×1280	1983 MB
GigaPath	1,134,953,984	223.42 G	270×1536	2382 MB
GigaPath-WSI	1,221,284,864	700.063.00 G	1×786	1.65 MB

Table 5. Details of models used by Yottixel to acquire patch embedding. GigaPath-WSI (the last row) is independent from Yottixel as it can generate a single embedding for WSI. Note that we used Yottixel’s “Bunch of Barcodes” (BoB) scheme to process and store embeddings. This enabled us to perform experiments faster and store embeddings much more efficiently.

This extensive training enables Virchow to generate highly expressive embeddings for WSI tiles, suitable for large-scale image retrieval tasks in pathology.

GigaPath is a vision transformer designed for pretraining foundation models on gigapixel WSIs⁸. It uses a two-stage self-supervised learning approach tailored for ultra-large histopathology datasets. Initially, the model performs image-level pretraining using DINOv2, focusing on learning localized tissue features. In the second stage, *GigaPath* progresses to whole-slide-level pretraining, combining a masked autoencoder with the LongNet architecture, which is optimized for ultra-large-context modeling. *GigaPath* was pretrained on 1.3 billion 256×256 image tiles extracted from 171,189 pathology slides, representing over 30,000 patients from 28 cancer centers within the Providence health network. This diverse dataset spans 31 tissue types, significantly larger than previous datasets such as The Cancer Genome Atlas (TCGA). *GigaPath*’s ability to capture both local and global features from gigapixel slides enables it to generate powerful WSI embeddings, making it effective for WSI-to-WSI retrieval tasks. Table 5 provides general information about the models used.

Discussions

This study evaluated the performance of several foundation models for zero-shot WSI retrieval in histopathology, revealing both potential benefits and limitations. While foundation models like UNI, Virchow, and *GigaPath* demonstrated improvements over traditional models such as DenseNet, which were trained on more general domains, significant challenges remain in achieving clinical-grade retrieval accuracy. The average F1 score of approximately 44% for zero-shot WSI retrieval tasks, as reported in this study, illustrates that current models are not yet suitable for clinical application.

A key finding from the results is the variability in retrieval performance across different organs. Models such as UNI performed well for homogeneous tissues like the kidneys and bladder, which present more uniform histological patterns that are easier for patch-based embeddings to capture. In contrast, for more complex and heterogeneous organs, such as the lungs and cervix, retrieval performance dropped significantly. This variability suggests that while foundation models are effective in specific contexts, they struggle to generalize across the diverse tissue structures found in histopathology. The organ-specific retrieval performance, shown in Tables 2 and 3, and 4, highlights this discrepancy, particularly in organs with high morphological variability, where local patches may not provide sufficient contextual information for meaningful retrieval.

The low retrieval performance for certain organs underscores a key limitation of patch-based retrieval methods. Although computationally efficient, patch-based approaches may overlook important spatial relationships and global tissue architectures that are critical for accurate diagnosis in complex organs. These limitations are especially pronounced in tissues where local morphological features vary greatly, further highlighting the need for embedding methods that capture both local and global features of whole-slide images (WSIs). This point aligns with findings from previous studies, which also reported limited improvements from aggregation techniques^{9,10}.

Another important observation from this study is the limited success of WSI-level aggregation methods, such as those used in *GigaPath* WSI. Aggregating patch embeddings into a single WSI-level vector theoretically simplifies retrieval by reducing the number of comparisons, but the results indicate that this approach often leads to a loss of critical spatial information. For instance, in cases like adrenal glands and head and neck tissues, *GigaPath* WSI underperformed compared to patch-based approaches, as shown in Tables 2 and 3, and 4. These results suggest that the aggregation process may obscure the finer details necessary for accurate retrieval, confirming that representing an entire WSI with a single vector remains a significant challenge in computational pathology^{1,11}. While aggregation has shown marginal improvements in certain cases, the overall benefits are minimal, further emphasizing the importance of patch-based methods for certain applications².

These findings highlight a central challenge in computational pathology: balancing computational efficiency with retrieval accuracy. Aggregation techniques, while reducing the storage and computational demands of WSI retrieval, may come at the cost of accuracy, particularly for tissues that require detailed spatial representation. This trade-off suggests that future research should explore hybrid approaches, combining the strengths of both patch-based and WSI-level embeddings to improve retrieval outcomes. Such approaches could address the limitations of existing methods by providing more holistic representations of WSI data.

The zero-shot retrieval setting used in this study presents an additional layer of complexity. Unlike fine-tuned models, which are trained on specific datasets and tasks, zero-shot models rely solely on pre-trained embeddings to generalize across unseen data. While the foundation models evaluated here demonstrated promise in zero-

shot settings, particularly in organs with simpler histological structures, their overall performance remains below the threshold required for clinical application. The average F1 scores for top-5 retrievals, which hover around 40% for most models, are far from achieving the accuracy necessary for supporting real-world diagnostic workflows.

This limitation is particularly evident in histopathology, where the consequences of misclassification or poor retrieval can directly impact patient outcomes. The zero-shot retrieval approach provides an efficient method for evaluating model generalization, but it is clear that fine-tuning on domain-specific datasets will be necessary to enhance retrieval accuracy and meet clinical standards. This aligns with previous research, which has noted that the order of accuracy improvement through aggregation is relatively low, further reinforcing the need for domain-specific refinement^{9,10}.

Given the limitations observed in this study, future research should focus on several key areas to improve the applicability of foundation models for histopathology image retrieval. First, there is a need for the development of new WSI-level embedding techniques that better capture the global context of WSIs. Hybrid models that combine patch-based features with global WSI representations could enhance retrieval accuracy by capturing both local and contextual information. Second, fine-tuning foundation models on organ-specific datasets may help address the organ variability seen in this study. Tailoring models to specific tissues, particularly those with complex structures, could improve retrieval performance in organs where the current models struggle. Third, more sophisticated aggregation techniques that retain critical spatial information while reducing computational complexity will be essential for making WSI retrieval systems more practical for large-scale clinical use. Finally, while this study provides a strong foundation for understanding the limitations of current models, clinical validation studies are necessary to assess the real-world utility of these models in supporting diagnostic workflows.

Overall, advanced foundation models like UNI and GigaPath consistently outperformed the baseline DenseNet model, but WSI-level retrieval accuracy remains low for most organs, particularly those with more complex structures. Retrieval performance was highly organ-dependent, with models showing better results for homogeneous tissues like the kidneys but struggling with complex organs like the lungs. The aggregation method employed by GigaPath did not significantly improve performance, indicating that important spatial information might be lost during the aggregation process. The zero-shot setting presented additional challenges, with foundation models falling short of the clinical standards required for accurate retrieval. These findings underscore the need for further advancements in both model design and retrieval methodologies to bring zero-shot WSI retrieval closer to clinical applicability.

Data availability

The datasets used/analyzed during the current study are available in “The Cancer Genome Atlas” repository (TCGA), <https://portal.gdc.cancer.gov/>.

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Author contributions

SA run most experiments, provided results and edited the manuscript. GA contributed to many experiments and data preparation. SH, KRK contributed to the design of experiments and analyzed results. JJG provided histopathology guidance. HRT designed the overall experiments, data requirements, and wrote most parts of the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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